Vishal Balaji

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Summary

Cloud / platform engineer with 3+ years at BMW Group Munich, building and operating **Kubernetes** platforms (**AWS EKS** and bare metal) with **Terraform**, **GitHub Actions CI/CD** and **Grafana** / **Prometheus** observability. Cut infra spend **~87%** across **5 plants** at **~99.9% uptime**. **German C1** and English (Professional Working Proficiency).

Experience

BMW Group, Integration Specialist (Autonomous Driving in Plant) — Munich, Germany
Apr 2024 to present

- Build and operate the **Kubernetes** platform (**AWS EKS** and bare metal) running Java/ Spring Boot microservices with **AWS RDS** and **Kafka**, linking plant floor hardware to cloud.
- Migrated 5 Java services from VMs to **Terraform** managed **Kubernetes** across **5 plants**, cutting infra cost from **EUR 15k to ~EUR 2k/yr (~87%)** with zero downtime deployments at **~99.9% uptime**.
- Built **GitHub Actions CI/CD** with custom Java metrics, **Grafana** dashboards and **Prometheus** / CloudWatch centralised log analysis.
- Cut service cold start from 60s to **<10s** by containerising with Quarkus; added HA to a UDP service via ingress load balancing.
- Own production on call and incident resolution; **mentored 3 interns**, led **2 external contractors**.

BMW Group, SpeedUp Scholarship / Internships (Robotics and Perception) — Munich, Germany
Sept 2021 to Jan 2024

- Robotics thesis and internship (grade **1.0**): NVIDIA Omniverse Digital Twins for industrial 6D pose estimation.
- LiDAR 3D detection on Waymo Open Dataset (**TensorFlow 2**, PointPillars) and YOLOv5 + TensorRT for plant floor autonomy.

CARISSMA Research Institute at TH Ingolstadt, Student Programming Assistant (HiWi) — Ingolstadt, Germany
Mar 2021 to July 2023

- Traffic camera detection and tracking: YOLOv5 + SORT + TensorRT on an **89k image** dataset; open sourced (19 stars, 5 forks).

Skills

Cloud and Platform: AWS (EKS, S3, CloudWatch), Kubernetes, Docker, Terraform, GitHub Actions CI/CD, Prometheus, Grafana, Linux, SLURM

Backend and Scripting: Python, Java, Spring Boot, Quarkus, Microservices, REST APIs, Kafka, AWS RDS, Git

Local AI Stack: Self hosts LLMs on a 2x GPU home system with llama.cpp and LangGraph agentic flows

Languages

German: C1 (Professional Working Proficiency) | English: Professional Working Proficiency

Education

Autonomous Vehicle Engineering (B.Eng), Technische Hochschule Ingolstadt (THI) — Ingolstadt, Germany
Oct 2020 to Feb 2024

- Grade 1.4 (German system) | Bachelor Thesis 1.0.
- Lead Perception Developer, Schanzer Racing Formula Student Electric (Driverless).

Certifications

AWS Certified Solutions Architect Associate (SAA) — Mar 2026